

[Request] how much water/energy (or may be compare the cost) is used in streaming a 3 hour 4K video vs 3 hour of chating with Ai (prompt can range from normal usages to image and videos generation as per the allowed limit for a base/max tier paying subscription)

Posted by u/Darkkiller059 • • 0 points • 73 comments

i found this idea dumb that both the think are same

if this thing would have been remotely true then we would have been in crisis way before AI

Comments

u/[deleted] • 1 points • 2026-03-28 02:01 UTC

[removed]

u/dagmarski • 1 points • 2026-03-27 23:52 UTC

Do most people in the comments miss the point? Streaming Netflix itself does not cost a lot of energy, however, displaying it on your tv does.

A somewhat energy efficient tv uses 50W. A single chatGPT prompt uses 0.34Wh. This means a single prompt is equivalent to 24 seconds of a tv. Not a lot?

u/[deleted] • 1 points • 2026-03-27 23:45 UTC

[removed]

u/JakSandrow • 2 points • 2026-03-27 21:05 UTC

there's also the difference that when i watch something on netflix at least I know that it's what i'm paying for. generative AI doesn't actually make anything new, it's just slop.

u/Elbobosan • 2 points • 2026-03-27 20:18 UTC

The whole water thing is stupid. If they built every data center proposed and retired none of the many many many that already exist the total still wouldn't be a significant fraction of the water used to grow Almonds in California, which is less than half of what the US uses to water its lawns.

Power, however, is a HUGE problem. It's somewhere between a 50-250% increase in the production capacity of the continental US. And we already needed to do something like that just to electrify transport. For reference not as a realistic power plant suggestion... Each acre of data center (1 big warehouse building) would need a solar farm the in the range of 150 acres of land. For commercial/light industrial it would be 2-8 acres.

The only thing that comes to mind as comparable to the density of AI power consumption is Aluminum smelting.

u/[deleted] • 1 points • 2026-03-27 20:05 UTC

[removed]

u/pangapingus • 2 points • 2026-03-27 19:15 UTC

I work in streaming and AI is def much more. Modern streaming as we know it is HLS/DASH where when you click on something to watch, you're sent a text-based .m3u8/.mpd file that includes any time-based ad break triggers and the series of segments (.ts files) to play in order, along with things like language, subtitles, display/audio settings, etc. Even the audio/video being played back are segmented chunks of thick media files, many video on-demand workflows leverage a conversion step in their pipeline to take the MP4 of a movie into a manifest (.m3u8/.mpd) and all segments (.ts) start-to-finish.

Meanwhile AI, especially as conversational context expands, is a much higher order level of compute and RAM needs; HLS/DASH can be stored in standard non-RAM storage, accessed once via a CDN, cached in the CDN, and just given out from cache at the edge over HTTP(S) from then on. AI meanwhile, interactions are still stateless over HTTP(S) but the backend isn't just stored file fetching, it's live compute on the AI provider's side, and it is very intensive. Even streaming workflows that use Server Side Ad Insertion (SSAI) is minimal compute in comparison, where once your manifest hits an ad break, hits the streaming server, which then reaches out to an Ad Decision Server (ADS), and it's largely just a text-based XML file of possible ads to play as the response, which then gets stitched into the stream.

tl;dr behind the scenes, streaming is just standard disk storage being simply fetched, AI requires intensive compute, RAM, and GPU power and as tasks/conversations expand the resource requirements continue to grow even further.

u/iicup2000 • 1 points • 2026-03-27 19:02 UTC

AI data centers use most water when training and testing new models. Already released models use very little water when prompting, so this question should be refocused from how much used when chatting with AI to how much used training AI

u/[deleted] • 1 points • 2026-03-27 18:01 UTC

[removed]

u/OverlyFriedEggs • 1 points • 2026-03-27 17:52 UTC

Im sure there's huge data centers for Netflix, amazon, hulu, etc, but just streaming movies doesn't take a lot of computing. I have a 1060 and a w2125 lenovo think station, and I mostly use it for plex and my own data storage. All my 4k movies run perfectly fine, the issue would be when hundreds of thousands of other people are also using that same services. Long story short, self host your life and your data. Its not hard

u/Hunterrcrafter • 8 points • 2026-03-27 17:01 UTC

Reacting to the video:

AI DATACENTERS DO NOT USE WATER, PEOPLE

They use energy, which is normally generated using fossil fuels, and those cause pollution.

Yes, some datacenters are watercooled, but that's either a closed loop, meaning water gets recirculated and doesn't get 'used' or 'consumed', or they use riverwater to cool and then discard the slightly warmer water, but still, the water doesn't get polluted or used up, just warmed up.

u/GoreyGopnik • 4 points • 2026-03-28 15:52 UTC

Some do. Many don't, and just use municipal water or natural water sources. In the former case, this makes the initially potable water need to be sent to wastewater facilities to be reprocessed, and in the latter case, this can lessen or exhaust the water in that source, which people often rely on.

Around 80% of the water used by datacenters (of which the growth of AI has caused a massive increase in construction) is evaporated, which lowers the amount of water that is actually drinkable to the public at any given time, and can lead to that water leaving the local ecosystem.

[<https://www.eesi.org/articles/view/data-centers-and-water-consumption>]

(<https://www.eesi.org/articles/view/data-centers-and-water-consumption>)

[<https://www.brookings.edu/articles/ai-data-centers-and-water/>]

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